

Mathematical Statistics Assignment 3

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3.2.5

3.2.5. Let $X \sim N(10, 25)$. Compute:

(a) $P(X \leq 20)$

(b) $P(X > 5)$

(c) $P(12 \leq X \leq 15)$

(d) $P(|X - 12| \leq 15)$

Solution

$$(a) \quad P(X \leq 20) = P\left(Z \leq \frac{20 - 10}{5}\right) = P(Z \leq 2) = 0.9772$$

$$(b) \quad P(X > 5) = P\left(Z > \frac{5 - 10}{5}\right) = P(Z > -1) = 0.8413$$

$$(c) \quad P(12 \leq X \leq 15) = P\left(\frac{12 - 10}{5} \leq Z \leq \frac{15 - 10}{5}\right)$$

$$= P(0.4 \leq Z \leq 1)$$

$$= P(Z \leq 1) - P(Z < 0.4)$$

$$= 0.8413 - 0.6554 = 0.1859$$

$$(d) \quad P(|X - 12| \leq 15) = P(-15 \leq X - 12 \leq 15) = P(-3 \leq X \leq 27)$$

$$= P\left(\frac{-3 - 10}{5} \leq Z \leq \frac{27 - 10}{5}\right)$$

$$= P(-2.6 \leq Z \leq 3.4)$$

$$= P(Z \leq 3.4) - P(Z < -2.6)$$

$$= 0.9997 - 0.0047 = 0.9950$$

3.2.15

The efficiency X of a certain electrical component may be assumed to be a random variable that is distributed uniformly between 0 and 100 units. What is the probability that X is:

- (a) Between 60 and 80 units?
- (b) Greater than 90 units?
- (c) Interpret (a) and (b).

Solution

The given problem involves a random variable X that is uniformly distributed between 0 and 100 units, $X \sim U(0, 100)$.

For a uniform distribution, the probability density function (PDF) is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Here, $a = 0$ and $b = 100$, so:

$$f(x) = \begin{cases} \frac{1}{100} & \text{for } 0 \leq x \leq 100 \\ 0 & \text{otherwise} \end{cases}$$

(a) Probability that X is between 60 and 80 units:

$$P(60 \leq X \leq 80) = \int_{60}^{80} f(x) dx$$

Since $f(x) = \frac{1}{100}$ for $0 \leq x \leq 100$:

$$P(60 \leq X \leq 80) = \int_{60}^{80} \frac{1}{100} dx = \frac{1}{100} \times (80 - 60) = \frac{1}{100} \times 20 = 0.2$$

(b) Probability that X is greater than 90 units:

$$P(X > 90) = \int_{90}^{100} f(x) dx$$

$$P(X > 90) = \int_{90}^{100} \frac{1}{100} dx = \frac{1}{100} \times (100 - 90) = \frac{1}{100} \times 10 = 0.1$$

(c) Interpretation:

1. **Between 60 and 80 units:** There is a 20% chance that the efficiency X of the component falls between 60 and 80 units.
2. **Greater than 90 units:** There is a 10% chance that the efficiency X of the component exceeds 90 units.

3.2.18

Suppose that the life length of a mechanical component is normally distributed.

- (a) If $\sigma = 3$ and $\mu = 100$, find the reliability of such a system at 105 h.
- (b) What should be the expected life of the component if it has reliability of 0.90 for 120 h?

Solution

Given:

- Mean (μ) = 100 hours
- Standard deviation (σ) = 3 hours
- The life length of the component follows a normal distribution.

(a) Find the reliability of the system at 105 hours.

The reliability at time t is given by:

$$R(t) = P(X > t) = 1 - P(X \leq t)$$

We convert this to a standard normal distribution using:

$$Z = \frac{X - \mu}{\sigma}$$

For $t = 105$:

$$Z = \frac{105 - 100}{3} = \frac{5}{3} \approx 1.67$$

Using the cumulative distribution function (CDF) of the standard normal distribution:

$$P(X \leq 105) = P(Z \leq 1.67)$$

From the standard normal table or using statistical software:

$$P(Z \leq 1.67) \approx 0.9525$$

Therefore:

$$R(105) = 1 - 0.9525 = 0.0475$$

Thus, the reliability of the system at 105 hours is approximately 0.0475.

(b) Find the expected life of the component if it has reliability of 0.90 for 120 hours.

Given $R(120) = 0.90$, we need to find the corresponding mean life μ .
The reliability $R(t)$ is:

$$R(t) = P(X > t) = 1 - P(Z \leq Z_t)$$

For $t = 120$:

$$P(Z \leq Z_t) = 1 - 0.90 = 0.10$$

From the standard normal table, the Z -value for which $P(Z \leq Z_t) = 0.10$ is:

$$Z_t \approx -1.28$$

Now, solving for μ :

$$Z_t = \frac{t - \mu}{\sigma}$$
$$-1.28 = \frac{120 - \mu}{3}$$

$$\mu = 120 + 1.28 \times 3 = 120 + 3.84 = 123.84$$

Thus, the expected life of the component for a reliability of 0.90 at 120 hours is approximately 123.84 hours.

3.2.21

A professor in a large statistics class has a grading policy such that only the 15% of the students with the highest scores will receive the grade A. The mean score for this class is 72 with a standard deviation of 6. Assuming that all the grades for this class follow a normal probability distribution, what is the minimum score that a student in this class has to get to receive an A grade?

Solution

We are asked to find the minimum score required to be in the top 15% of a statistics class where the scores are normally distributed with:

- Mean (μ) = 72
- Standard deviation (σ) = 6

Solution:

1. Define the problem:

We need to find the score X such that the top 15% of students have a score higher than X . This corresponds to finding the 85th percentile (since $100\% - 15\% = 85\%$) of the distribution.

2. Find the Z-score for the 85th percentile:

Using the standard normal distribution, we need the Z-score Z such that:

$$P(Z \leq Z_{0.85}) = 0.85$$

From standard normal tables or using statistical software, we find:

$$Z_{0.85} \approx 1.04$$

3. Convert the Z-score to an actual score:

The relationship between the Z-score and the score X in a normal distribution is given by:

$$Z = \frac{X - \mu}{\sigma}$$

Substituting $Z = 1.04$, $\mu = 72$, and $\sigma = 6$:

$$1.04 = \frac{X - 72}{6}$$

Solving for X :

$$X = 1.04 \times 6 + 72 = 6.24 + 72 = 78.24$$

Therefore, we can say that, the minimum score that a student needs to achieve to be in the top 15% of the class and receive an A grade is approximately **78.24**.

3.3.6

3.3.6. Find the marginal density functions of the random variables X and Y if their joint pdf is given by:

$$f(x, y) = \begin{cases} \frac{1}{5}(3x - y), & 1 \leq x \leq 2, 1 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Solution

To find the marginal density functions of the random variables X and Y , we integrate the joint probability density function $f(x, y)$ over the appropriate ranges.

Marginal density of X :

The marginal density function $f_X(x)$ is obtained by integrating the joint pdf $f(x, y)$ over all values of y :

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Given the support of $f(x, y)$ is $1 \leq x \leq 2$ and $1 \leq y \leq 3$, we integrate over y from 1 to 3:

$$f_X(x) = \int_1^3 \frac{1}{5}(3x - y) dy$$

Now compute the integral:

$$\begin{aligned} f_X(x) &= \frac{1}{5} \int_1^3 (3x - y) dy \\ &= \frac{1}{5} \left[3xy - \frac{y^2}{2} \right]_1^3 \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= \frac{1}{5} \left[\left(3x \cdot 3 - \frac{3^2}{2} \right) - \left(3x \cdot 1 - \frac{1^2}{2} \right) \right] \\ &= \frac{1}{5} \left[\left(9x - \frac{9}{2} \right) - \left(3x - \frac{1}{2} \right) \right] \\ &= \frac{1}{5} [6x - 4] \end{aligned}$$

Thus:

$$f_X(x) = \frac{6x - 4}{5}, \quad \text{for } 1 \leq x \leq 2$$

Marginal density of Y :

The marginal density function $f_Y(y)$ is obtained by integrating the joint pdf $f(x, y)$ over all values of x :

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Given the support of $f(x, y)$ is $1 \leq x \leq 2$ and $1 \leq y \leq 3$, we integrate over x from 1 to 2:

$$f_Y(y) = \int_1^2 \frac{1}{5}(3x - y) dx$$

Now compute the integral:

$$\begin{aligned} f_Y(y) &= \frac{1}{5} \int_1^2 (3x - y) dx \\ &= \frac{1}{5} \left[\frac{3x^2}{2} - yx \right]_1^2 \end{aligned}$$

Substituting the limits:

$$\begin{aligned}
&= \frac{1}{5} \left[\left(\frac{3 \cdot 2^2}{2} - 2y \right) - \left(\frac{3 \cdot 1^2}{2} - 1y \right) \right] \\
&= \frac{1}{5} \left[(6 - 2y) - \left(\frac{3}{2} - y \right) \right] \\
&= \frac{1}{5} \left[\frac{12}{2} - \frac{3}{2} - 2y + y \right] \\
&= \frac{1}{5} \left[\frac{9}{2} - y \right]
\end{aligned}$$

Thus:

$$f_Y(y) = \frac{9 - 2y}{10}, \quad \text{for } 1 \leq y \leq 3$$

Therefore, The marginal density functions are:

$$f_X(x) = \frac{6x - 4}{5}, \quad 1 \leq x \leq 2$$

$$f_Y(y) = \frac{9 - 2y}{10}, \quad 1 \leq y \leq 3$$

3.3.8

Find k so that $f(x, y) = kxy$, $1 \leq x \leq y \leq 2$, will be a pdf. Also find:

(i) $P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right)$

(ii) $P\left(X + Y \leq \frac{3}{2}\right)$

Solution

To solve the given problem, we proceed as follows:

Step 1: Finding k

For $f(x, y) = kxy$ to be a valid probability density function (pdf), the integral over the entire region must equal 1:

$$\int \int f(x, y) dx dy = 1$$

Given $f(x, y) = kxy$ for $1 \leq x \leq y \leq 2$, we set the limits of integration as x from 1 to y , and y from 1 to 2:

$$\int_1^2 \int_1^y kxy \, dx \, dy = 1$$

Step 1.1: Integrating with respect to x :

$$\begin{aligned} \int_1^2 \left(\int_1^y kxy \, dx \right) dy &= \int_1^2 k \left(\int_1^y xy \, dx \right) dy \\ &= \int_1^2 k \left[\frac{x^2}{2} \cdot y \right]_1^y dy \\ &= \int_1^2 k \left(\frac{y^3}{2} - \frac{y}{2} \right) dy \end{aligned}$$

Step 1.2: Integrating with respect to y :

$$\begin{aligned} &= \frac{k}{2} \int_1^2 (y^3 - y) \, dy \\ &= \frac{k}{2} \left[\frac{y^4}{4} - \frac{y^2}{2} \right]_1^2 \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= \frac{k}{2} \left(\left(\frac{16}{4} - \frac{4}{2} \right) - \left(\frac{1}{4} - \frac{1}{2} \right) \right) \\ &= \frac{k}{2} \left((4 - 2) + \frac{1}{4} \right) = \frac{k}{2} \times \frac{9}{4} \\ &= \frac{9k}{8} \end{aligned}$$

Setting this equal to 1:

$$\frac{9k}{8} = 1 \quad \Rightarrow \quad k = \frac{8}{9}$$

Step 2: Calculating the Probabilities

(i) $P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right)$

We integrate $f(x, y)$ over the region $1 \leq x \leq \frac{3}{2}$ and $1 \leq y \leq \frac{3}{2}$:

$$P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right) = \int_1^{\frac{3}{2}} \int_1^y \frac{8}{9} xy \, dx \, dy$$

Step 2.1: Integrating with respect to x :

$$\begin{aligned}
 &= \int_1^{\frac{3}{2}} \frac{8}{9} \left(\int_1^y xy \, dx \right) dy \\
 &= \int_1^{\frac{3}{2}} \frac{8}{9} y \left[\frac{x^2}{2} \right]_1^y dy \\
 &= \int_1^{\frac{3}{2}} \frac{8}{9} y \left(\frac{y^2}{2} - \frac{1}{2} \right) dy
 \end{aligned}$$

Step 2.2: Integrating with respect to y :

$$\begin{aligned}
 &= \frac{8}{9} \int_1^{\frac{3}{2}} \left(\frac{y^3}{2} - \frac{y}{2} \right) dy \\
 &= \frac{8}{9} \times \frac{1}{2} \int_1^{\frac{3}{2}} (y^3 - y) dy \\
 &= \frac{4}{9} \left[\frac{y^4}{4} - \frac{y^2}{2} \right]_1^{\frac{3}{2}}
 \end{aligned}$$

Substituting the limits:

$$= \frac{4}{9} \left(\left(\frac{\left(\frac{3}{2}\right)^4}{4} - \frac{\left(\frac{3}{2}\right)^2}{2} \right) - \left(\frac{1}{4} - \frac{1}{2} \right) \right)$$

Continue simplifying this expression to get the final probability.

(ii) $P(X + Y \leq \frac{3}{2})$

Since X and Y are both in the range $[1, 2]$, the sum $X + Y \geq 2$ is always greater than $\frac{3}{2}$. Thus:

$$P\left(X + Y \leq \frac{3}{2}\right) = 0$$

Final Answers:

- $k = \frac{8}{9}$
- $P(X \leq \frac{3}{2}, Y \leq \frac{3}{2})$ is calculated from the integral.
- $P(X + Y \leq \frac{3}{2}) = 0$.

3.3.9

The random variables X and Y have a joint pdf:

$$f(x, y) = \begin{cases} \frac{8}{9}xy, & 1 \leq x \leq y \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

TABLE 3.2 Joint pmf of X and Y .				
		Y		
X	-2	0	1	4
-1	0.3	0.1	0.0	0.2
3	0.0	0.2	0.1	0.0
5	0.1	0.0	0.0	0.0

Solution

(a) The marginal of X is:

$$f_X(x) = \int_x^2 f(x, y) dy = \int_x^2 \frac{8}{9}xy dy = \frac{4}{9}(4x - x^3), \quad 1 \leq x \leq 2.$$

(b) $P(1.5 < X < 1.75, Y > 1)$:

$$\begin{aligned} P(1.5 < X < 1.75, Y > 1) &= \int_{1.5}^{1.75} \left(\int_x^2 \frac{8}{9}xy dy \right) dx = \int_{1.5}^{1.75} \frac{4}{9}(4x - x^3) dx \\ &= \frac{4}{9} \left[2x^2 - \frac{x^4}{4} \right]_{1.5}^{1.75} = 0.2426. \end{aligned}$$

3.4.2

The joint pdf of (X, Y) is:

$$f(x, y) = \frac{1}{\theta^2} e^{-\frac{x+y}{\theta}}, \quad x, y > 0, \theta > 0.$$

Find the pdf of $U = X - Y$.

Solution

To find the PDF of $U = X - Y$, given the joint PDF of (X, Y) :

$$f(x, y) = \frac{1}{\theta^2} e^{-\frac{x+y}{\theta}}, \quad x, y > 0, \quad \theta > 0,$$

we proceed as follows:

Step 1: Define the Transformation

Let $U = X - Y$ and $V = Y$. The inverse transformation is:

$$X = U + V, \quad Y = V$$

Step 2: Jacobian of the Transformation

The Jacobian determinant for the transformation from (X, Y) to (U, V) is:

$$\frac{\partial(X, Y)}{\partial(U, V)} = \begin{vmatrix} \frac{\partial X}{\partial U} & \frac{\partial X}{\partial V} \\ \frac{\partial Y}{\partial U} & \frac{\partial Y}{\partial V} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

Step 3: Joint PDF of U and V

Since the Jacobian determinant is 1, the joint PDF of (U, V) is:

$$f_{U,V}(u, v) = f_{X,Y}(u + v, v) \times |1|$$

Substitute $f_{X,Y}(x, y)$:

$$f_{U,V}(u, v) = \frac{1}{\theta^2} e^{-\frac{(u+v)+v}{\theta}} = \frac{1}{\theta^2} e^{-\frac{u+2v}{\theta}}$$

for $u + v > 0$ and $v > 0$.

Step 4: Marginal PDF of U

To find the marginal PDF of U , integrate out V :

$$f_U(u) = \int_0^\infty f_{U,V}(u, v) dv$$

Substitute the expression for $f_{U,V}(u, v)$:

$$f_U(u) = \int_0^\infty \frac{1}{\theta^2} e^{-\frac{u+2v}{\theta}} dv$$

Factor out terms that do not depend on v :

$$f_U(u) = \frac{1}{\theta^2} e^{-\frac{u}{\theta}} \int_0^\infty e^{-\frac{2v}{\theta}} dv$$

Now, compute the integral:

$$\int_0^\infty e^{-\frac{2v}{\theta}} dv = \left[-\frac{\theta}{2} e^{-\frac{2v}{\theta}} \right]_0^\infty$$

At $v = \infty$, $e^{-\frac{2v}{\theta}} \rightarrow 0$, and at $v = 0$, $e^{-\frac{2 \cdot 0}{\theta}} = 1$:

$$\int_0^\infty e^{-\frac{2v}{\theta}} dv = \frac{\theta}{2}$$

Substitute back:

$$f_U(u) = \frac{1}{\theta^2} e^{-\frac{u}{\theta}} \times \frac{\theta}{2} = \frac{1}{2\theta} e^{-\frac{u}{\theta}}$$

Step 5: Final PDF of U

The PDF of $U = X - Y$ is:

$$f_U(u) = \frac{1}{2\theta} e^{-\frac{u}{\theta}}, \quad u > 0$$

3.5.4

Suppose that the number of cars arriving at a busy intersection in a given 20-min interval in a large city has a Poisson distribution with mean 120. Determine a lower bound for the probability that the number of cars arriving in a given 20-min period will be between 100 and 140 using Chebyshev's inequality.

Solution

The number of cars arriving at a busy intersection in a given 20-minute interval follows a Poisson distribution with a mean of $\mu = 120$. We are asked to determine a lower bound for the probability that the number of cars arriving in a given 20-minute period will be between 100 and 140 using Chebyshev's inequality.

Step-by-Step Solution:

1. Understanding Chebyshev's Inequality: Chebyshev's inequality states:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

where μ is the mean, σ is the standard deviation, and $k > 0$.

The complementary probability is:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

This inequality gives a lower bound for the probability that X lies within k standard deviations of the mean. 2. Calculating the Standard Deviation: For a Poisson distribution:

$$\sigma^2 = \mu$$

$$\sigma = \sqrt{\mu} = \sqrt{120}$$

3. Identifying the Interval and Calculating k : We are asked for the probability that the number of cars is between 100 and 140.

$$P(100 \leq X \leq 140) = P(|X - 120| < 20)$$

Here, $|X - 120| < 20$, so the deviation from the mean is 20. Now calculate k :

$$k = \frac{20}{\sigma} = \frac{20}{\sqrt{120}} \approx \frac{20}{10.95} \approx 1.83$$

4. Applying Chebyshev's Inequality: Using Chebyshev's inequality:

$$P(|X - 120| < 20) \geq 1 - \frac{1}{k^2}$$

Substitute $k \approx 1.83$:

$$P(|X - 120| < 20) \geq 1 - \frac{1}{(1.83)^2}$$

$$P(|X - 120| < 20) \geq 1 - \frac{1}{3.3489}$$

$$P(|X - 120| < 20) \geq 1 - 0.2987 \approx 0.7013$$

Final Answer: Using Chebyshev's inequality, the lower bound for the probability that the number of cars arriving in a 20-minute period is between 100 and 140 is approximately 0.7013.

3.5.6

How large should the size of a random sample be so that we can be 90% certain that the sample mean $\bar{X}$ will not deviate from the true mean by more than $\sigma/2$?

Solution

To solve this problem, we need to determine the sample size n required to ensure that the sample mean $\bar{X}$ is within $\frac{\sigma}{2}$ of the true mean with 90% confidence.

Step-by-Step:

1. Confidence Interval for the Sample Mean: The confidence interval for the sample mean $\bar{X}$ is given by:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where:

- $Z_{\alpha/2}$ is the critical value from the standard normal distribution for a 90% confidence level.
- σ is the population standard deviation.
- n is the sample size.

2. Critical Value: For a 90% confidence level, $Z_{\alpha/2}$ corresponds to the 95th percentile of the standard normal distribution because:

$$\alpha = 1 - 0.90 = 0.10 \quad \text{and} \quad \frac{\alpha}{2} = 0.05$$

From standard normal tables or using statistical software:

$$Z_{0.05} \approx 1.645$$

3. Margin of Error: We are given that the margin of error E should be no more than $\frac{\sigma}{2}$, so:

$$E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \frac{\sigma}{2}$$

4. Solve for n : Set the margin of error equal to $\frac{\sigma}{2}$:

$$1.645 \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{2}$$

Now solve for n :

$$\frac{1.645}{\sqrt{n}} = \frac{1}{2}$$

$$\sqrt{n} = 1.645 \times 2 = 3.29$$

$$n = 3.29^2 \approx 10.8241$$

Since the sample size n must be an integer, we round up to the next whole number:

$$n = 11$$

Final Answer: The required sample size is $n = 11$.

3.5.13

Suppose that 2500 customers subscribe to a telephone exchange. There are 80 trunk lines available. Any one customer has the probability of 0.03 of needing a trunk line on a given call. Consider the situation as 2500 trials with probability of “success” $p = 0.03$. What is the approximate probability that the 2500 customers will “tie up” the 80 trunk lines at any given time?

Solution

To solve this problem, we can use the Poisson approximation to the Binomial distribution. Here’s the step-by-step solution:

Step 1: Identify the given parameters

- Number of customers (trials), $n = 2500$
- Probability of needing a trunk line on a given call (success), $p = 0.03$
- Number of available trunk lines, $k = 80$

Step 2: Approximating with Poisson distribution When n is large and p is small, the binomial distribution $\text{Binomial}(n, p)$ can be approximated by a Poisson distribution with mean $\lambda = np$.

Calculate λ :

$$\lambda = n \times p = 2500 \times 0.03 = 75$$

Thus, we use $\text{Poisson}(75)$ to approximate the number of customers needing trunk lines at a given time.

Step 3: Calculate the probability We want to find the probability that at least 80 customers need trunk lines simultaneously, i.e., $P(X \geq 80)$ where $X \sim \text{Poisson}(75)$.

Using the complement rule:

$$P(X \geq 80) = 1 - P(X \leq 79)$$

Step 4: Compute $P(X \leq 79)$ The cumulative probability $P(X \leq 79)$ can be computed using statistical software or Poisson distribution tables.

Using a Poisson cumulative distribution function (CDF) from statistical software or a calculator:

$$P(X \leq 79) \approx 0.8166$$

Step 5: Find $P(X \geq 80)$ Now, calculate $P(X \geq 80)$:

$$P(X \geq 80) = 1 - P(X \leq 79)$$

$$P(X \geq 80) = 1 - 0.8166 = 0.1834$$

Final Answer:

The approximate probability that the 2500 customers will "tie up" the 80 trunk lines at any given time is 0.1834, or 18.34%.

Projects for Chapter 3

3A Mixture distribution

In statistical modeling, if the data are contaminated by outliers or if the samples are drawn from a population formed by a mixture of two populations, one could use mixture distributions. Mixture distributions are used frequently in medical applications, such as microarray analysis. Suppose a random variable X has pdf $f_1(x)$ with probability p_1 and pdf $f_2(x)$ with probability p_2 , where $p_1 + p_2 = 1$. Then we say that the random variable X has a *mixture distribution*. This can be thought of as observing a Bernoulli random variable Z that is equal to 1 with probability p_1 and 2 with probability p_2 . Thus,

$$X = \begin{cases} X_1 \sim f_1(x), & \text{if } Y = 1, \\ X_2 \sim f_2(x), & \text{if } Y = 2. \end{cases}$$

- (a) Show that the pdf of X is given by $f(x) = p_1 f_1(x) + p_2 f_2(x)$.
- (b) If (μ_1, σ_1^2) and (μ_2, σ_2^2) are means and variances of $f_1(x)$ and $f_2(x)$, respectively, show that

$$\mu = E(X) = p_1 \mu_1 + p_2 \mu_2,$$

and

$$\sigma^2 = \text{Var}(X) = p_1 \sigma_1^2 + p_2 \sigma_2^2 + p_1 \mu_1^2 + p_2 \mu_2^2 - (p_1 \mu_1 + p_2 \mu_2)^2.$$

Solution

Solution

Part (a): Show that the pdf of X is given by $f(x) = p_1 f_1(x) + p_2 f_2(x)$

Since X is drawn from one of two distributions, depending on the outcome of a Bernoulli random variable Z , the overall probability density function (pdf) of X can be described as a mixture of the two individual pdfs, weighted by their respective probabilities.

- If $Z = 1$, $X \sim f_1(x)$ with probability p_1 .
- If $Z = 2$, $X \sim f_2(x)$ with probability p_2 .

The total pdf $f(x)$ is the weighted sum of the two pdfs:

$$f(x) = P(Z = 1)f_1(x) + P(Z = 2)f_2(x)$$

Substituting $P(Z = 1) = p_1$ and $P(Z = 2) = p_2$:

$$f(x) = p_1f_1(x) + p_2f_2(x)$$

Thus, the pdf of X is $f(x) = p_1f_1(x) + p_2f_2(x)$.

Part (b): Show that $\mu = E(X) = p_1\mu_1 + p_2\mu_2$ and $\sigma^2 = \text{Var}(X) = p_1\sigma_1^2 + p_2\sigma_2^2 + p_1\mu_1^2 + p_2\mu_2^2 - (p_1\mu_1 + p_2\mu_2)^2$

Step 1: Mean of X

The expected value of X is the weighted sum of the expected values of $f_1(x)$ and $f_2(x)$:

$$\mu = E(X) = E(X|Z = 1)P(Z = 1) + E(X|Z = 2)P(Z = 2)$$

Substituting the expected values and probabilities:

$$\mu = \mu_1p_1 + \mu_2p_2$$

Thus, $\mu = p_1\mu_1 + p_2\mu_2$.

Step 2: Variance of X

The variance of X is given by:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

We calculate $E[X^2]$ first:

$$E[X^2] = E[X^2|Z = 1]P(Z = 1) + E[X^2|Z = 2]P(Z = 2)$$

Now, $E[X^2|Z = 1] = \sigma_1^2 + \mu_1^2$ and $E[X^2|Z = 2] = \sigma_2^2 + \mu_2^2$:

$$E[X^2] = (\sigma_1^2 + \mu_1^2)p_1 + (\sigma_2^2 + \mu_2^2)p_2$$

Now calculate $\text{Var}(X)$:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Substituting $E[X^2]$ and $E[X]$:

$$\text{Var}(X) = p_1(\sigma_1^2 + \mu_1^2) + p_2(\sigma_2^2 + \mu_2^2) - (p_1\mu_1 + p_2\mu_2)^2$$

Expanding the terms:

$$\text{Var}(X) = p_1\sigma_1^2 + p_2\sigma_2^2 + p_1\mu_1^2 + p_2\mu_2^2 - (p_1\mu_1 + p_2\mu_2)^2$$

Thus:

$$\sigma^2 = \text{Var}(X) = p_1\sigma_1^2 + p_2\sigma_2^2 + p_1\mu_1^2 + p_2\mu_2^2 - (p_1\mu_1 + p_2\mu_2)^2$$

3F Generating a large amount of random data using R

Although Projects 3B and 3E can be used for random sample generation, it is tedious for large amounts of data generation. Now we will give some R commands for generating n random samples from a few common distributions.

`runif(n, min=a, max=b)` will generate n random values from $U(a, b)$. For instance, to generate 100 random samples from uniform $U(0, 2)$, use the command `runif(100, 0, 2)`.

Similarly, `rnorm(n, mean=a, sd=b)` will generate n random values from normal distribution with mean a and standard deviation b . Thus, `rnorm(100, mean=4, sd=2)` will generate 100 random values from a normal distribution with mean 4 and standard deviation 2. Similarly, `rbinom(n, m, p)` will generate n random values from a binomial distribution with parameters m and p . For instance, to generate 100 sample values from binomial with $m = 10$, and $p = 0.5$, use `rbinom(100, 10, 0.5)`. Similarly, `rexp(n, rate)` will generate n random samples from an exponential distribution with a specified rate. Thus, `rexp(100, 1/1000)` will generate 100 exponential values with rate $\lambda = 1/1000$.

Similarly, we can generate random samples from any distribution.

Exercise 3F. For each of the generated random samples, write an R-code to create a histogram overlapped by the corresponding density function.

Solution

Python Code for Generating Random Samples and Plotting

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import uniform, norm, binom, expon

# Set the number of samples
n = 100

# Generate and plot for uniform distribution U(0, 2)
uniform_samples = np.random.uniform(0, 2, n)
sns.histplot(uniform_samples, kde=True, stat="density", bins=10, color="skyblue", label="Histogram")
x = np.linspace(0, 2, 1000)
plt.plot(x, uniform.pdf(x, loc=0, scale=2), 'r-', label="Uniform PDF")
plt.title("Uniform Distribution U(0, 2)")
plt.legend()
plt.show()

# Generate and plot for normal distribution N(4, 2^2)
normal_samples = np.random.normal(4, 2, n)
sns.histplot(normal_samples, kde=True, stat="density", bins=10, color="skyblue", label="Histogram")
x = np.linspace(-2, 10, 1000)
plt.plot(x, norm.pdf(x, loc=4, scale=2), 'r-', label="Normal PDF")
plt.title("Normal Distribution N(4, 2^2)")
plt.legend()
plt.show()

# Generate and plot for binomial distribution Binomial(10, 0.5)
binomial_samples = np.random.binomial(10, 0.5, n)
sns.histplot(binomial_samples, kde=False, stat="density", bins=10, color="skyblue", label="Histogram")
x = np.arange(0, 11)
plt.plot(x, binom.pmf(x, 10, 0.5), 'r-', marker="o", label="Binomial PMF")
plt.title("Binomial Distribution Binomial(10, 0.5)")
plt.legend()
plt.show()

# Generate and plot for exponential distribution Exp(1/1000)
```

```

exponential_samples = np.random.exponential(1000, n)
sns.histplot(exponential_samples, kde=True, stat="density",
             bins=10, color="skyblue", label="Histogram")
x = np.linspace(0, 5000, 1000)
plt.plot(x, expon.pdf(x, scale=1000), 'r-', label="
Exponential PDF")
plt.title("Exponential Distribution Exp(1/1000)")
plt.legend()
plt.show()

```

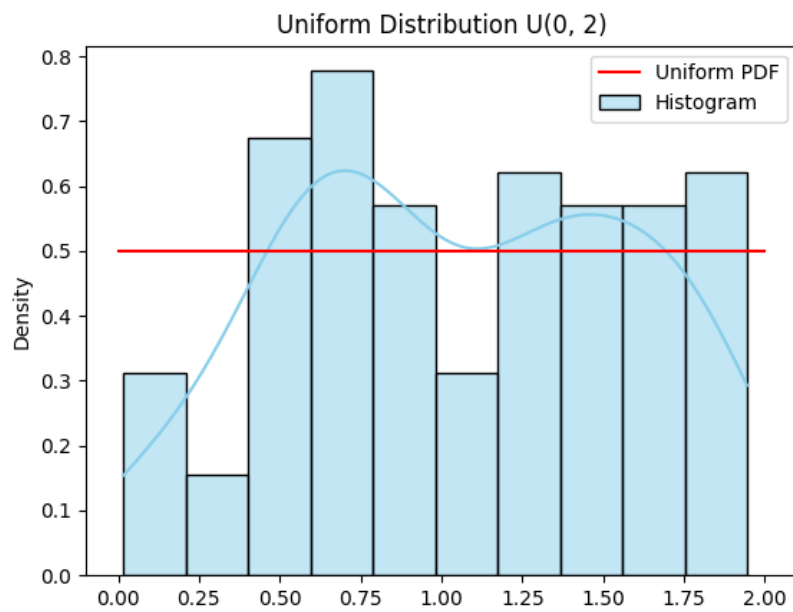


Figure 1: Table - Uniform Distribution.

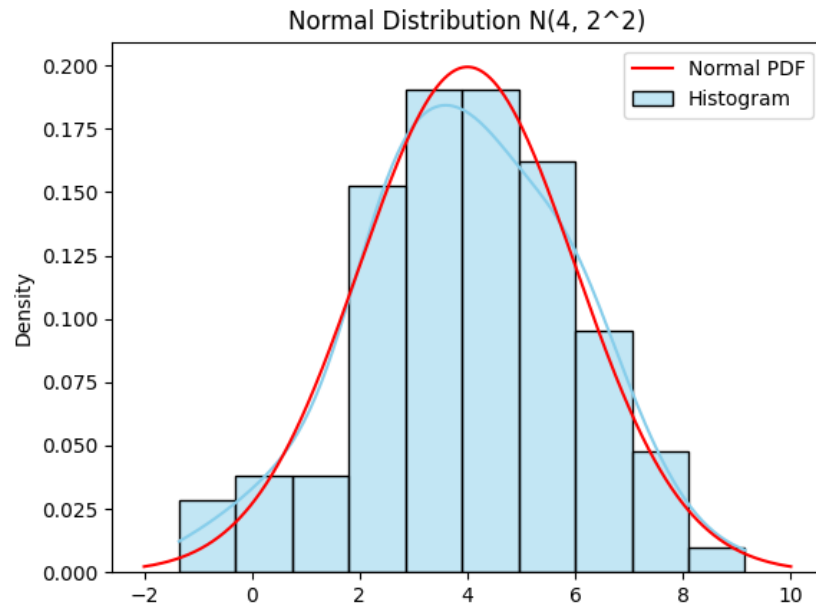


Figure 2: Table - Normal Distribution.

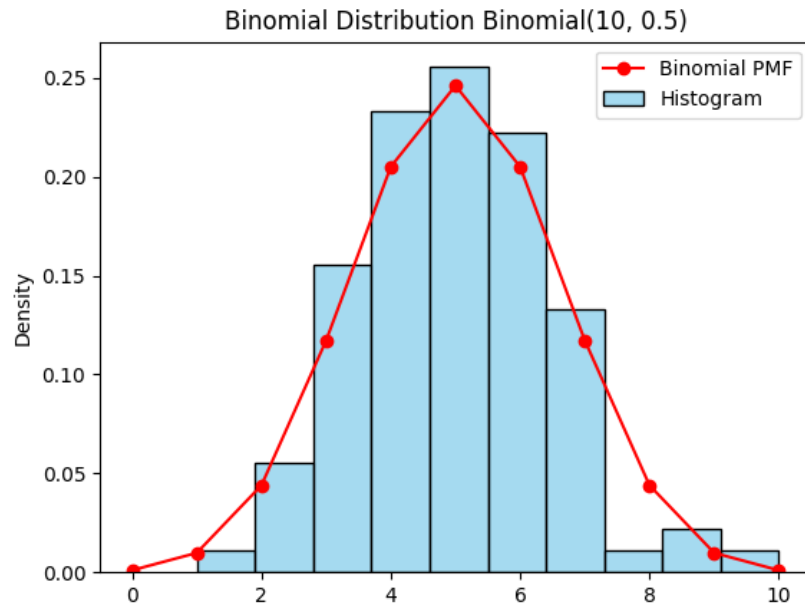


Figure 3: Table - Binomial Distribution.

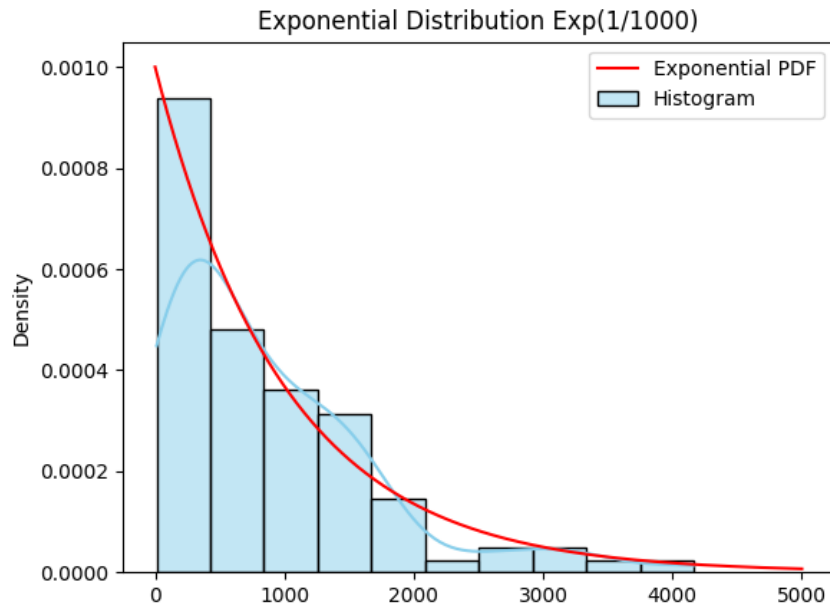


Figure 4: Table - Exponential Distribution.